

MURALI KRISHNA N

Data Scientist | Machine Learning Engineer | GenAI & Agentic AI Developer

+91-9901669235 | muralikrishna22ds@gmail.com | Bengaluru, India

<https://www.linkedin.com/in/murali-krishna-n-ds/> | github.com/MuraliKrishna22ds | Available for Immediate Joining

PROFESSIONAL SUMMARY

Results-driven Data Scientist with 6+ months of hands-on experience building end-to-end ML pipelines and extracting actionable insights from large-scale datasets. Proven expertise in NLP text classification, customer segmentation, and Generative AI / RAG applications across healthcare, airlines, and retail. Proficient in Python, SQL, scikit-learn, XGBoost, LangChain, Azure OpenAI, and MLflow with a strong track record in model evaluation, SHAP explainability, and stakeholder-ready reporting. Targeting Data Scientist, ML Engineer, or Gen AI / Agentic AI Developer roles in Bengaluru or Hyderabad.

CORE COMPETENCIES

Languages	Python 3, SQL (MySQL),
Libraries	Pandas, NumPy, scikit-learn, TensorFlow, XGBoost, LightGBM, SHAP, Matplotlib, Seaborn
Machine Learning	Supervised & Unsupervised Learning, Classification, Regression, Clustering, Ensemble Methods, Hyperparameter Tuning, Cross-Validation, PCA
NLP & GenAI	NLP, TF-IDF, n-grams, Transformers, LLMs, Prompt Engineering, RAG, Agentic AI, Embeddings, Vector Stores
LLM / GenAI Tools	LangChain, Mixtral-8x7B, Azure OpenAI, Azure AI Search, Multi-Agent Architecture, REST API
MLOps & Tools	MLflow, Batch Scoring, Data Drift Monitoring, Feature Engineering, EDA, ETL Pipelines, Git, Power BI, Jupyter

PROFESSIONAL EXPERIENCE

Data Scientist Intern | Intenso Tech Solution Pvt. Ltd., Bengaluru

Sep 2025 – Mar 2026

- Engineered end-to-end Python and SQL data pipelines processing 50,000+ records; resolved missing values, outliers, and applied feature encoding — reducing data inconsistencies by ~20%.
- Built classification and regression models using scikit-learn and XGBoost, achieving F1-score of 0.87+ and ROC-AUC of 0.91+ through systematic hyperparameter tuning and stratified cross-validation.
- Designed domain-specific features using PCA-based dimensionality reduction and correlation analysis, improving model signal quality and predictive accuracy.
- Delivered data visualizations using Matplotlib, Seaborn, and Power BI dashboards, translating complex model outputs into stakeholder-ready business insights.
- Logged 12+ algorithm experiment runs with hyperparameters and evaluation metrics in MLflow, enabling reproducible benchmarking and faster model iteration cycles.

KEY PROJECTS

Adverse Event Seriousness Classification

Client: Nova Pharm, USA | Domain: Healthcare / Clinical Safety Tech: Python, scikit-learn, XGBoost, SQL, MLflow, SHAP, NLP (TF-IDF, n-grams), Pandas

- Built a supervised ML text classifier to categorize adverse event narratives by seriousness level, reducing manual pharmacovigilance review time and accelerating regulatory submission workflows.
- Extracted TF-IDF and n-gram features from clinical text; joined with SQL-sourced structured case data to create a hybrid feature matrix, improving classification accuracy on imbalanced datasets.
- Applied stratified cross-validation, class-weight tuning, and threshold optimization to maximize recall on the minority (serious) class, minimizing false negatives in patient safety decisions.
- Generated SHAP explainability reports for regulatory reviewers, surfacing top predictive features aligned with FDA documentation requirements.
- Deployed automated batch scoring pipelines with data drift detection; versioned all models and experiments in MLflow for full reproducibility.

GenAI Operations & Policy Assistant (RAG)

Client: AeroLink Airways, APAC | Domain: Airlines **Tech:** Python, Azure OpenAI, LangChain, Azure AI Search (Vector), Embeddings, Prompt Engineering

- Developed a production-ready Generative AI RAG assistant enabling airline operations staff to retrieve policy guidance and SOP information with grounded, cited answers — significantly reducing manual lookup time.
- Ingested and indexed operational manuals using optimized chunking strategies tailored for procedural and tabular content, improving retrieval relevance and coverage.
- Designed prompt templates generating concise, cited responses with confidence-based fallback logic to handle out-of-scope queries gracefully.
- Implemented hybrid retrieval with metadata filters for improved domain-specific search across IRROPS, baggage, and rebooking workflows.
- Optimized system latency and cost through targeted prompt tuning; validated results using scenario-based evaluation datasets.

Customer Segmentation – Unsupervised Machine Learning

Domain: Retail Analytics | Personal Project **Tech:** Python, scikit-learn, K-Means, DBSCAN, PCA, Matplotlib, Seaborn, SQL, Power BI

- Applied K-Means and DBSCAN clustering on 30,000+ retail customer records; used SQL for data extraction and PCA for 2D visualization — identified 5 distinct actionable customer personas.
- Improved marketing campaign targeting efficiency by ~18% through segment-based insights; delivered a Power BI dashboard communicating cluster profiles and recommended business actions.
- Evaluated cluster quality using Silhouette Score and Davies-Bouldin Index; iterated on feature selection to maximize inter-cluster separation and intra-cluster cohesion.

EDUCATION

B.E. – Artificial Intelligence & Machine Learning

2021 – 2025

Jyothy Institute of Technology, Karnataka, India | **CGPA: 8.1 / 10**

CERTIFICATIONS & ACHIEVEMENTS

- B.E. in Artificial Intelligence & Machine Learning — CGPA 8.1/10 | Jyothy Institute of Technology, 2025
- 6-Month Data Science Internship Certificate — Intenso Tech Solution Pvt. Ltd., Bengaluru (Python, SQL, ML, EDA, MLflow, Power BI)
- Delivered 3 end-to-end AI/ML projects: clinical NLP text classification (Nova Pharm, USA), GenAI RAG policy assistant (AeroLink Airways, APAC), and unsupervised retail customer segmentation.
- Proficient in SHAP model explainability, MLflow experiment tracking, production batch scoring pipelines, and automated data drift monitoring.